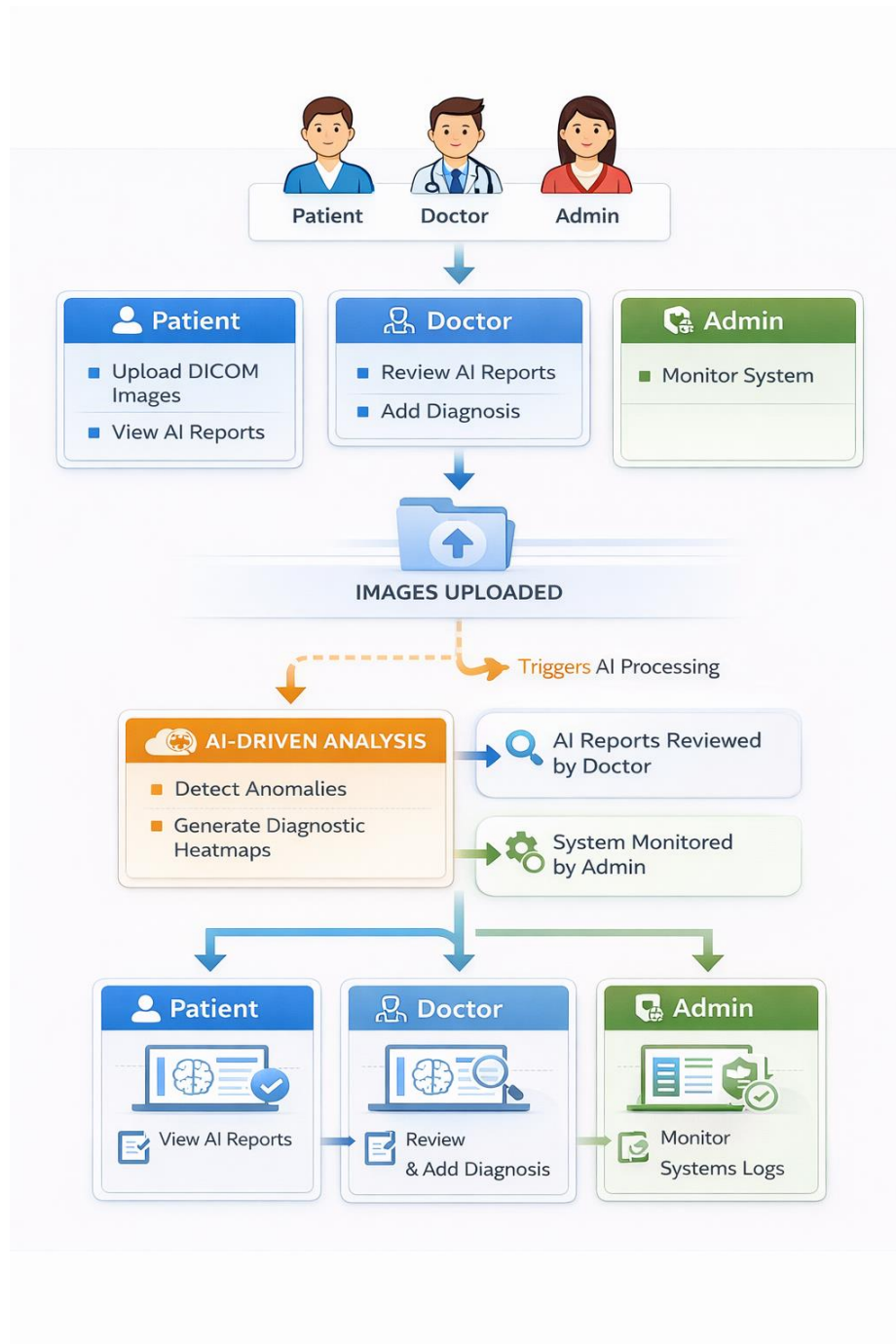


# Project Title : CuraMind AI – HIPAA-Compliant Telehealth & Diagnostics Platform

Date: April 16, 2026



A brief overview of the application

# 1. Introduction

## 1.1 Purpose

This document defines the requirements for **CuraMind AI**, a secure SaaS platform bridging patients and medical specialists. It utilizes Computer Vision to pre-process high-resolution medical images (MRI, X-ray) to assist radiologists by highlighting anomalies.

## 1.2 Scope

CuraMind AI provides secure patient-doctor interaction, DICOM image handling, AI-assisted diagnostic support, and Electronic Medical Record (EMR) management. The system ensures data privacy through role-based access and auditability for healthcare compliance.

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# 2. Overall Description

## 2.1 Product Perspective

CuraMind AI is a standalone web application designed with a monolithic architecture to simplify initial HIPAA compliance.

## 2.2 Tech Stack

- **Backend:** Django (Monolith) for built-in security and mature ORM.
- **Asynchronous Tasks:** Celery + Redis/RabbitMQ for non-blocking image processing.
- **Database:** PostgreSQL for transactional integrity and ACID compliance.
- **Image Handling:** pydicom for parsing medical metadata.
- **Storage:** AWS S3 with private ACLs for secure, compliant hosting.

## 2.3 User Classes

- **Patient:** Can register, upload DICOM images, view records, and book appointments.
- **Doctor:** Can access patient data, review AI-processed images, and add diagnoses.
- **Admin:** Manages users and monitors system compliance logs.

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## 3. System Features & Functional Requirements

### 3.1 Security & Access Control (RBAC)

- **Requirement:** Restrict access based on user roles (e.g., Nurses cannot view Doctor-specific data).
- **Implementation:** Leverage Django's permission system and django-guardian for object-level permissions.

### 3.2 Medical Image Pipeline

- **Requirement:** Support for high-resolution DICOM files with metadata extraction.
- **Implementation:** Secure file uploads with MIME-type checking and metadata stripping.

### 3.3 AI-Driven Diagnostic Support

- **Requirement:** Automated analysis to highlight anomalies.
- **Implementation:** Celery workers pass images to a pre-trained CV model (e.g., ResNet) to generate diagnostic heatmaps.

### 3.4 Audit Logging

- **Requirement:** Log every critical action (view, edit, delete, access attempt) on a medical record.
  - **Implementation:** Immutable, time-stamped tables via dedicated logging middleware/signals.
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## 4. Non-Functional Requirements

### 4.1 Security & Compliance

- Strict adherence to HIPAA/GDPR standards.
- Encryption at rest and SSL/TLS for data in transit.

### 4.2 Performance

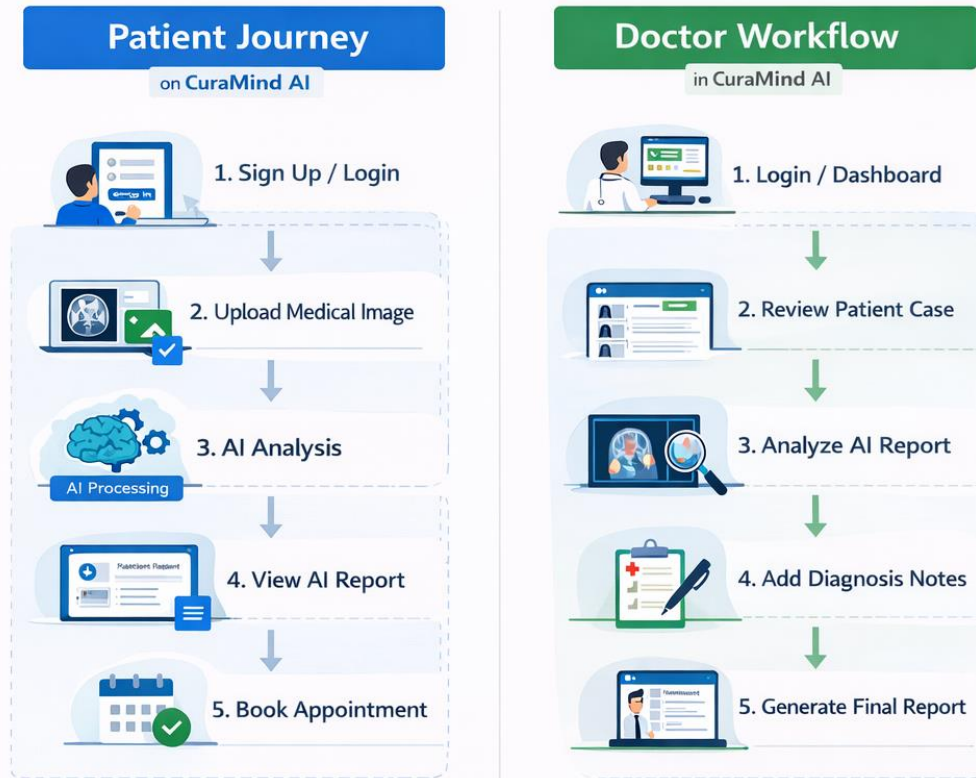
- Main web thread must remain responsive during peak image processing loads.
- AI analysis must be handled asynchronously via Celery.

### 4.3 Reliability & Quality

- **Unit Tests:** Target 100% (minimum >90%) coverage.
- **Penetration Testing:** Mandatory testing to attempt bypassing permissions.

## CuraMind AI

### Patient Journey in CuraMind AI



### Security Layer: Applies to both journeys



## Patient Dashboard



Dashboard

Upload Images

Reports

My Appointments

Settings

CuraMind AI

Search patients...



**Welcome, John Doe!**  
Manage your medical images and appointments.

 Upload New Medical Image

**Recent Reports**

 AI Diagnostic Report  
**X-Ray - Chest**  

Ready

View

 AI Diagnostic Report  
**MRI - Brain**  

95%

View

View AI Reports

**Upcoming Appointment**

**Dr. Sarah Miller**  
Cardiologist  
Today 209 A24 10:30 AM

**Jane Smith**  
Female  
Tomorr09 A8r 10:30 AM

View Details >



EMAIL

.....

Forget Password ?

**SIGN IN** >

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Bone Specialist - King Hospital

**DR . John sunny**  
Heart Specialist - Omega Hospital

**DR . Ritual**  
Child Specialist - Mama care Hospital

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Child Specialist - Mama care Hospital

**About**  
DR. Ritual is one of the most experienced and qualified child specialists. She has 8 years of working experience and worked at Civil Hospital. Currently, she is practicing at Asar memorial hospital medical center.

**BOOK AN APPOINTMENT**

< February 2021 >

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